

Jaspreet Singh Nahal

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

SUMMARY

AI Developer specializing in LLMs, RAG Systems, and FastAPI, with internship experience and **400+ LeetCode problems**. Proven track record building conversational AI, trading systems, and automation tools using Python, React, FastAPI, and LangChain with focus on scalable architecture and performance optimization.

EXPERIENCE

Machine Learning Intern

Aug 2025 – Oct 2025

Expedien eSolutions

Noida, India

- Built RAG-based legal query resolver using LangChain, FAISS, and Hugging Face models with RetrievalQA pipeline
- Optimized document retrieval achieving <2s response time on 10K+ document corpus with Streamlit UI

Algorithmic Trading Developer

Nov 2024 – Dec 2024

Freelance

Remote

- Developed automated trading system with SMA crossover strategy using Flask API and PostgreSQL
- Reduced trade execution latency by 30% through async processing and optimized data handling

PROJECTS

EverydayGPT | [Live Demo](#) | *PyTorch, Transformers, FAISS, RAG, NLP*

Feb 2026

- Designed and pretrained a 205M-parameter GPT-style Transformer from scratch in PyTorch with causal self-attention, custom initialization, and GPT-2 tokenization on a 10B-token FineWeb-Edu + Wikipedia corpus.
- Built a Retrieval-Augmented Generation (RAG) pipeline using SentenceTransformers and FAISS, achieving sub-205ms retrieval latency and reducing hallucinations via confidence-based gating.
- Implemented instruction fine-tuning, selective loss masking, top-k sampling, and repetition control for stable multi-turn conversational performance.

CodeAtlas | *Python, FastAPI, NetworkX, RAG, TypeScript*

Mar 2026

- Built a full-stack code intelligence platform that ingests GitHub repos/ZIPs/code snippets and generates interactive multi-language dependency graphs.
- Implemented async FastAPI job processing and a filterable React force-graph UI to highlight architecture domains, key files, and dependency cycles in large repositories.

AI Assistant with NER | *React.js, Python, Flask, MongoDB*

Sep 2024

- Developed NLP conversational assistant with 92% intent recognition accuracy handling 1,000+ user sessions
- Integrated Named Entity Recognition and real-time Flask API with <1s response time

EDUCATION

B.Tech in Computer Science Engineering

2023 – 2027

Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript

ML & LLM Systems: PyTorch, Hugging Face Transformers, SentenceTransformers, FAISS, LangChain, Retrieval-Augmented Generation (RAG)

Data & Training: NumPy, Pandas, scikit-learn, Spark, gradient-based optimization, large-scale pretraining

Backend & Tooling: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, Git, Docker, GitHub Actions, Streamlit

HACKATHONS & CERTIFICATIONS

Competitive Achievements: 1st Place – Code Farming (643 participants), DSA Championship (24-hr contest, 2026); 3rd Place – Voyager Blitz, HackIndia Spark-4 (HackerRank, Mar 2026)

Hackathons: IBM AI Hackathon (2025), Google Solutions Challenge (2025), GUVI Hackathon (2025)

Certifications: Google – Prompt Design in Vertex AI & Responsible AI; IBM – Accelerating Deep Learning with GPUs; BCG X (Forage) – GenAI Financial Chatbot Simulation